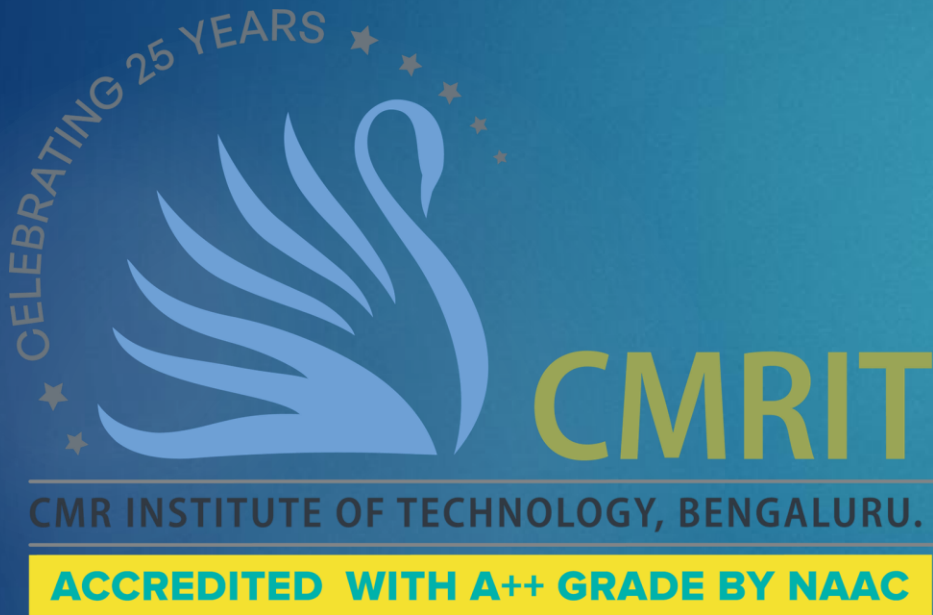


Smartguard Snatch Theft Detection.



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Problem Statement

Violent & Fast

Snatch thefts happen in seconds — victims have no time to react or press buttons during violent attacks

Infrastructure Dependent

Existing CCTV and GPS systems require fixed infrastructure or internet connectivity — not personal or portable

No Affordable Solution

No automatic personal safety device exists in the Indian market that works offline and is accessible to everyone

Existing Solutions & Their Failures

Current approaches cannot address violent snatch thefts effectively

Solution	How It Works	Why It Fails
CCTV Systems	Fixed cameras monitor areas	Infrastructure dependent, not personal
Manual Panic Buttons	Victim presses button to alert	Cannot press during violent snatch
Gesture Wearables	Detects specific hand gestures	Needs deliberate movement, impractical
Vehicle GPS Trackers	Tracks car location only	Only for vehicles, not personal use

Literature Survey

Research reviewed across three critical domains to identify gaps that SmartGuard addresses

Theft Detection Systems

YOLO-based CCTV detection (Chandrika et al, 2020) and CNN pose estimation (Ramesh & Priya, 2021) require fixed infrastructure — not wearable or portable

Wearable Safety Devices

GPS-GSM panic buttons (Sharmila et al, 2019) require manual activation. Fall detection (Kavitha & Mohan, 2021) only detects falls, not theft

Accelerometer & GSM Systems

Mathie et al (2004) validated 2G threshold for involuntary movement. Dernbach et al (2012) achieved 96% accuracy with Random Forest classification

Identified Research Gaps



Infrastructure Dependent

Theft detection systems cannot be worn personally — require fixed cameras or indoor installations



Manual Activation Required

Wearable safety devices need victim input during violent attack — impossible when grabbed



Never Applied to Theft

Accelerometer detection validated for falls and activity recognition — never adapted for theft scenarios



Vehicle-Only Systems

GSM-GPS alert systems exist for vehicles but never adapted for personal anti-theft wearable applications

SmartGuard — Bridging The Gap

Factor	Existing Systems	SmartGuard
Activation	Manual	Automatic
Connectivity	Internet needed	GSM only, works anywhere
Portability	Fixed or bulky	Wearable, clips to bag
Response Time	Minutes	Under 3 seconds
Cost	High	Under ₹2200

What is SmartGuard?

A smart wearable that automatically detects a snatch theft and instantly alerts your emergency contact with your live location — no human intervention needed

Universal Attachment

Attaches to any bag, backpack, or purse using secure clip mechanism

Works Offline

GSM network connectivity means no internet required — works anywhere in India

Instant Alert

Alerts sent within 3 seconds of detection with exact GPS location

Hardware Components

Arduino Uno	Main microcontroller — brain of system	₹400-600
MPU6050	Detects sudden violent pull via G-force spike	₹99-200
SIM800L	Sends SMS alert to emergency contact	₹350-500
Neo-6M GPS	Captures exact location coordinates	₹350-500
Piezo Buzzer	Triggers loud alarm to scare attacker	₹20-50
Li-ion Battery	Portable power supply	₹150-200
Breadboard & Wires	Circuit connections	₹100-150

Total Cost: Under ₹2200

How SmartGuard Works

Wear It

Person attaches SmartGuard to bag

Monitor

MPU6050 continuously monitors acceleration

Alarm

Buzzer triggers immediately — scares attacker

Locate

GPS captures exact coordinates

Detect

Violent snatch detected — G-force spike crosses threshold

Alert

SMS sent to emergency contact

Respond

Map link received within 3 seconds



Fully Automatic

Zero input needed from victim during attack



Works Offline

GSM network, no internet required



Wearable

Clips onto any bag, discreet and hidden



Instant Alert

SMS delivered in under 3 seconds



Affordable

Entire system under ₹2200



Scalable

Adaptable for children, elderly, vehicles

Future Scope

Police Integration

Direct police station integration via API

Miniaturization

Keychain size using Arduino Nano

Camera Module

Capture attacker's face

Mobile App

Alert management and history

SmartGuard — Because safety shouldn't depend on your reaction time"